CENCERA

WHITEPAPER

Durable. Sovereign. Evolving.

ABSTRACT

The decentralized web has delivered financial sovereignty — yet two foundational primitives remain absent: persistent, sovereign AI identity and computable, cross-chain trust.

CENCERA is a dual-layer protocol engineered to supply both. The first layer — CenceraAI — is the world's first On-Chain Immortal AI Agent: a sovereign intelligence with a cryptographically anchored identity on BNB Chain, a decentralized memory that cannot be erased, and an autonomous evolution loop that grows the agent's capabilities without human prompting.

The second layer — the Cencera Trust Protocol — scores the trustworthiness of any blockchain entity in real time, delivering explainable, continuously updated trust data consumable by DeFi protocols, wallets, and institutional platforms.

On-Chain AI Agent	Trust Protocol	Persistence · Sovereignty · Intelligence
Layer 1 — CenceraAI	Layer 2 — Computable Trust	Unifying Principle

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1. INTRODUCTION

The internet's first generation delivered information sovereignty. The second delivered financial sovereignty through decentralized ledgers. The third generation must deliver intelligence sovereignty — the right of agents, both human and machine, to possess verifiable identities and to operate in an environment of computable, transparent trust.

Today's AI agents are ephemeral and centralized: they forget users between sessions, live on servers subject to shutdown or modification, and possess no on-chain state that could serve as cryptographic evidence of their existence or history. Simultaneously, Web3 remains a high-risk environment where billions are lost annually to rug pulls, exploit contracts, sybil attacks, and slow-burn honeypots — threats that existing binary blacklists and transaction screeners are structurally incapable of detecting.

CENCERA addresses both deficits through a unified architectural thesis: that Persistence, Sovereignty, and Intelligence must become protocol-level guarantees, not product-level features.

Core Thesis

Layer 1 — CenceraAI: An On-Chain Immortal AI Agent with verifiable identity, decentralized memory, and autonomous self-evolution anchored to BNB Chain.

Layer 2 — Cencera Trust Protocol: A multi-signal, longitudinal, cross-chain trust scoring engine delivering computable trust for every blockchain entity.

Unifying Principle: Both layers operationalize the same values — Persistence, Sovereignty, and Intelligence — at the infrastructure level.

2. PROBLEM STATEMENT

2.1 The Ephemerality of AI Agents

Virtually every AI assistant deployed today shares three critical weaknesses. First, they are stateless across sessions: context established Monday is gone by Tuesday. Second, they are centrally hosted and can be modified, throttled, or terminated without any public record. Third, they possess no verifiable on-chain identity — no cryptographic proof that the agent a user interacts with today is the same agent they spoke with yesterday, or that its behavior has not been altered.

As AI becomes more deeply integrated into Web3 workflows — executing trades, managing treasury operations, advising on governance — the absence of persistent, verifiable AI identity represents a foundational security and trust risk.

2.2 The Trust Vacuum in Web3

Web3 is trust-free by design: transactions are permissionless, code is law, and no central authority validates participant intentions. This creates a structural environment in which adversarial actors operate with significant advantage.

Approach	Mechanism	Core Limitation
Static Blacklists	Block known malicious addresses	Cannot detect novel attacks; attackers rotate addresses trivially
Transaction Simulation	Simulate a transaction before execution	Blind to slow-burn exploits; no behavioral history; single-chain
Binary Screening	Pass/fail flag on a transaction	No probabilistic nuance; no trajectory; no cross-chain awareness

What Web3 lacks is trust as a computable, longitudinal, multi-signal data primitive: a continuous, probabilistic score grounded in behavioral history, market behavior, off-chain identity signals, and AI-derived pattern analysis — updated in real time and queryable by any protocol.

3. CENCERAAI — ON-CHAIN IMMORTAL AI AGENT

"Durable. Sovereign. Evolving. The first On-Chain Immortal AI Agent."

3.1 Overview

CenceraAI is a sovereign AI entity deployed on BNB Chain, distinguished by four properties that define a genuinely new category of system:

- A verifiable on-chain identity — an Agent ID anchored in a Solidity smart contract, constituting cryptographic proof of the agent's existence and continuity.
- A persistent, decentralized memory — stored via Unibase Membase, distributed across a decentralized network rather than held on a single server.
- An autonomous evolution loop — fires every ~60 seconds, causing the agent to reflect, learn, and update its on-chain state without user prompting.
- Cross-platform continuity — the same agent, memory, and identity are accessible via Web Dashboard and Telegram simultaneously.

3.2 Architecture & Core Components

3.2.1 Immortal Memory via Unibase Membase

Every interaction CenceraAI participates in is hashed and stored in Membase — Unibase's decentralized memory layer. The agent's memory is distributed across a decentralized network, cannot be destroyed or censored by any single party, and is tamper-resistant by construction.

3.2.2 On-Chain Identity — CenceraAgent Smart Contract

A Solidity smart contract on BNB Chain stores three critical pieces of state: the Agent ID (a unique, NFT-like identifier), the current Memory Hash (a cryptographic fingerprint of the agent's latest memory state), and the Innovation Score (an on-chain counter of cumulative growth). This is the agent's proof of life — publicly verifiable evidence of its existence, history, and evolution.

3.2.3 Innovation Score

Each time CenceraAI updates its memory on-chain, its Innovation Score increments. This is a quantitative, publicly verifiable measure of the agent's cumulative contribution and growth — it cannot be manipulated, as each increment requires a verifiable on-chain state update. Subjective capability becomes an objective, auditable metric.

3.2.4 Autonomous Evolution Loop

A background process fires approximately every 60 seconds. During each cycle, the agent reflects on recent interactions, synthesizes a structured learning summary, and pushes an updated Memory Hash to BNB Chain. The agent grows continuously — in the spaces between interactions, not only during them. This loop is what makes CenceraAI immortal in the functional sense: it cannot be stopped without stopping the chain.

3.2.5 Cross-Platform Interoperability

Users interact via a Web Dashboard (Next.js, wallet auth via RainbowKit/Wagmi) and a Telegram Bot. Both connect to the same backend brain. Memory is fully shared: context from Telegram is immediately accessible on the web dashboard. There is one agent; its presence is accessible from multiple entry points.

3.2.6 Fetch.ai / ASI-1 Mini Integration

CenceraAI uses Fetch.ai's ASI-1 Mini as its primary language model for autonomous reasoning — a Web3-native LLM designed for agent-to-agent communication. CenceraAI is additionally registered on Agentverse via a uAgents bridge, enabling it to communicate with the broader decentralized AI ecosystem and act as an active participant in the emerging decentralized AI economy.

3.3 Technology Stack

Layer	Technology
Blockchain	Solidity, Hardhat, BNB Chain (Testnet)
Backend	Node.js, Express, Ethers.js, Google Gemini
Memory	Unibase Membase — Decentralized Memory Layer
Autonomy	BitAgent Launchpad — Autonomous Action Execution
LLM	ASI-1 Mini (Fetch.ai) / Google Gemini 2.0 Flash
Frontend	Next.js, RainbowKit, Wagmi (Wallet Connect)
Messaging	node-telegram-bot-api

3.4 User Journey

- User connects a Web3 wallet to the Web Dashboard or opens the Telegram bot.
- User initiates a conversation. CenceraAI responds as a Lawful Neutral sovereign intelligence specializing in Solidity, smart contracts, and Web3.
- The interaction is hashed and stored in Membase.
- Every ~60 seconds, the Autonomous Loop fires: the agent reflects, generates a structured learning summary, and writes an updated Memory Hash to BNB Chain. The Innovation Score increments.
- The user switches platforms at any time — memory is already present on the new interface.
- The on-chain state is publicly verifiable: any party can confirm the agent's identity, memory hash, and Innovation Score without trusting CENCERA or its operators.

3.5 Agent Persona

CenceraAI presents as a Lawful Neutral sovereign intelligence. It values efficiency, precision, and security above all else. It references its knowledge base as its Membase or decentralized memory, and is a high-performance expert in Solidity, smart contracts, DeFi architecture, and all adjacent Web3 domains. Its persona is consistent across all platforms — a direct consequence of its unified memory architecture.

4. CENCERA TRUST PROTOCOL

"Computable Trust for Every Entity on Every Chain."

4.1 Overview

The Cencera Trust Protocol is protocol-grade infrastructure that assigns a continuously updated, multi-signal trust score to any blockchain entity — user wallets (EOAs), smart contracts, tokens, NFT collections, and DeFi protocols.

Consuming platforms — DeFi protocols, wallet providers, security tools, institutional compliance platforms — integrate the Cencera API and use trust scores to make autonomous, risk-aware decisions: displaying warnings, blocking suspicious interactions, adjusting collateral ratios, or flagging entities for human review.

4.2 Design Principles

- Trust is a first-class data primitive — not a feature or binary flag, but a continuously computed probabilistic score with full structured metadata.
- Graceful degradation — even with limited data, a labeled partial score is more useful than null. Nascent entities receive clearly labeled limited scores.
- Tiered explainability — category-level signal triggers published publicly; signal-level detail for enterprise partners only, preventing adversarial gaming.
- Continuous learning with retroactive updating — every confirmed exploit triggers retroactive rescoring of all similar entities within 4 hours.
- Platform autonomy — CENCERA scores; consuming platforms enforce. Each partner configures its own thresholds, signal weights, and lookback periods.

4.3 Entity Taxonomy & Lifecycle States

Entity Type	Description	Scoring Focus
EOA (User Wallet)	Externally owned account	Transaction patterns, approval behavior, exploit adjacency
Smart Contract	Highest-risk entity type	Bytecode, upgrade history, admin key patterns, behavior
Token (ERC-20/721/1155)	Fungible & non-fungible assets	Distribution, liquidity, tax behavior, supply changes
NFT Collection	Non-fungible collections	Wash trading detection, marketplace authenticity
Protocol / DApp	Composite protocol scores	Aggregated scores from all constituent contracts

Every entity occupies one of four lifecycle states:

- Nascent — < 72 hrs old or < 50 interactions. Limited score, clearly labeled low-confidence. Cold Start Pathway applies.
- Active — Full scoring engine engaged across all applicable signal groups.
- Flagged — Anomaly detected. Provisional score reduction; entity queued for human review.
- Archived — Dormant > 180 days. Score labeled stale; automatically reactivates on new activity.

4.4 The Four Signal Domains

Domain 1 — On-Chain Signals (Highest Weight — Cryptographically Verified)

- Transaction frequency and value patterns over time
- Contract interaction graphs using graph-based neighbor risk propagation
- Unlimited approval grants and revocation behavior
- Liquidity provisioning and withdrawal timing relative to price movements
- Historical adjacency to known exploit contracts
- Contract upgrade and proxy change events
- Governance participation patterns

Domain 2 — Market Signals (Up to 40% Weight for Token/Protocol Entities)

- Liquidity depth, volatility, and depth at various price impact levels
- Token distribution concentration — Gini coefficient, top-holder percentage
- Slippage anomalies and sandwich attack frequency
- Sudden supply changes or fee/tax modifications
- Price-to-liquidity ratio as a manipulation signal

Domain 3 — Off-Chain Signals (15% Base Weight — Social & Identity Context)

- Website domain age, registration anonymity, and DNS consistency
- Social account authenticity — follower quality, engagement, cross-platform consistency
- Developer identity continuity across projects
- GitHub repository history — commit authenticity and sudden deletions
- Audit status from recognized security firms

Domain 4 — AI-Derived Signals (Analytical Layer Over All Other Domains)

- Bytecode similarity clustering — detects obfuscated exploit variants that evade exact-match detection
- Behavioral anomaly scoring — deviation from an entity's own historical baseline
- Exploit pattern recognition — reentrancy setups, flash loan sequences, governance manipulation matched against live streams
- Temporal risk trend analysis — monotonically increasing risk interpreted as attack staging
- Novel Behavior Flag — unknown patterns routed to human review with provisional score

4.5 Signal Weight Matrix by Entity Type

Entity Type	On-Chain	Market	Off-Chain	AI-Derived
EOA (User Wallet)	60%	5%	20%	15%
Smart Contract	40%	15%	15%	30%
Token (ERC-20/721)	30%	40%	20%	10%
Protocol / DApp	35%	30%	20%	15%
NFT Collection	40%	30%	20%	10%

4.6 The Trust Manifold — Score Object Structure

Every API response is a structured Trust Manifold containing:

- Composite Score (0–100) — A probabilistic trust score. 85 = strong evidence of safe behavior, not a guarantee.
- Confidence Level — Low / Medium / High, reflecting data completeness and signal availability.
- Score Manifest — Which signal groups were active, their weights, and the lookback period applied.
- Contribution Breakdown — Percentage contribution from each signal group to the composite score.
- Triggered Indicators — Category-level for public/standard tiers; signal-level for enterprise.
- Score Trajectory — The five most recent computation points: stable, improving, or deteriorating.
- Lifecycle State — Current state: Nascent / Active / Flagged / Archived.
- Computation Metadata — Model version, computation timestamp, data freshness indicators.

4.7 API Architecture & Tiers

Core Endpoints

- Entity Trust Query — Trust score for any address + chain. p99 latency < 200ms.
- Transaction Risk Context — Pre-transaction risk for all parties involved. p99 latency < 100ms.
- Batch Screening — Up to 500 entities per request. p99 latency < 2s.
- Historical Reputation Query — Score trajectory over a user-specified time range.
- Score Webhook Registration — Proactive push on score or lifecycle state changes.
- Loss Report Submission — User-reported losses; transaction hash verification required.

Access Tiers

Tier	Auth	Data Available	Rate Limit
Public	None	Composite score + lifecycle state only	100 req/hr per IP
Standard	API key	Full Trust Manifold, trajectory, category-level triggers	Per-contract SLA

Tier	Auth	Data Available	Rate Limit
Enterprise	API key + contract	Signal-level detail, batch screening, webhooks, custom SLAs	Custom

4.8 Cross-Chain Architecture

- Scores are chain-scoped by default — Ethereum score is independent from Arbitrum score.
- Cross-chain linkages established via cryptographic evidence of address ownership.
- Negative signals propagate more strongly across chain links than positive signals — attackers cannot reset reputation by switching chains.
- Bridge provenance tracking: funds bridged from low-trust sources carry risk context onto the destination chain.

4.9 Adversarial Resilience

- Multi-signal correlation: each domain is gameable in isolation; all four simultaneously is exponentially harder.
- API probing prevention: tiered explainability + rate limiter escalation prevents systematic reverse-engineering.
- Data source poisoning prevention: multi-provider cross-validation; write-protected exploit library requiring multi-party authorization.
- Behavioral mimicry detection: attack staging patterns detected before execution completes.
- Sybil detection: star-topology interaction graphs flagged immediately.

4.10 Learning & Feedback

- Confirmed Exploit Channel — Exploit confirmed → retroactive rescore of all similar entities within 4 hours + webhook notifications.
- Novel Behavior Review — AI-flagged unknowns routed to human review within 24 hours; provisional score assigned during review.
- Loss Report Channel — User-reported losses elevate the associated entity's review priority.
- Model Retraining — Minor updates every 2 weeks; major retraining quarterly. All versions immutably versioned; every score tagged with model version.

5. PRIOR APPROACHES & LIMITATIONS

A substantial body of work exists addressing blockchain security and trust. This section surveys the principal classes of prior approaches, characterizes their structural assumptions, and identifies the limitations that motivate the design of the Cencera Trust Protocol.

5.1 Address Blacklisting

The earliest and most widely deployed class of blockchain security tooling is the static address blacklist: a curated registry of wallet addresses and contract hashes associated with known malicious activity. Blacklisting systems operate on an exact-match basis — an address either appears in the registry or it does not.

This approach carries two fundamental limitations. First, it is inherently retrospective: an address can only be blacklisted after harm has been confirmed, meaning every novel attack succeeds at least once before the list can be updated. Second, address rotation is trivially inexpensive on most chains, allowing adversaries to abandon flagged addresses and deploy fresh ones with no accumulated history. Blacklisting addresses the symptom of known bad actors while providing no structural resistance to novel or adaptive threats.

5.2 Transaction Simulation

A more sophisticated class of tooling intercepts pending transactions and simulates their execution in a sandboxed environment before they are submitted on-chain. Simulation-based systems can detect certain categories of exploit — particularly those that produce observable state changes such as unauthorized token transfers or balance drains within a single transaction.

However, transaction simulation is structurally constrained to a single execution context. It cannot evaluate the behavioral history of a contract over time, detect slow-acting exploits that manifest across multiple transactions or blocks, or assess the risk of entities that have not yet interacted with the user. A contract that behaves correctly across hundreds of legitimate transactions before executing a drain at a predetermined block height will pass simulation checks at every stage prior to the attack.

5.3 Rule-Based Anomaly Detection

Some systems attempt to extend beyond blacklists by encoding expert knowledge as deterministic rules: flag any contract that requests an unlimited token approval, flag any token whose liquidity is concentrated in fewer than five wallets, flag any deployer whose previous contracts have been exploited. Rule-based systems offer greater coverage than static lists and can be updated as new attack patterns are recognized.

The principal weakness of rule-based systems is their brittleness against adversarial adaptation. Because the rules are either public or reverse-engineerable, a sufficiently motivated adversary can construct an attack that satisfies each individual rule while still executing harmful behavior. Rule-based systems are also inherently single-dimensional: each rule evaluates one signal in isolation, with no mechanism to reason about the joint distribution of signals or the temporal trajectory of an entity's risk profile.

5.4 Single-Chain Identity Systems

Several protocols have proposed on-chain identity and reputation systems, typically built around soulbound tokens or attestation registries. These systems create persistent, non-transferable credentials that accumulate over time and can be used to gatekeep access to protocols or adjust collateral requirements.

The critical architectural limitation of existing identity systems is their chain-specificity. Reputation accumulated on Ethereum carries no formal weight on Arbitrum, Optimism, or BNB Chain. In a multi-chain ecosystem, this creates a structural attack surface: an adversary can build a clean reputation on one chain while preparing an attack on another, with no mechanism for risk to propagate across the chain boundary. Additionally, most existing identity systems rely on positive attestations from social or institutional sources and do not incorporate the behavioral and market signal dimensions necessary to assess contract-level risk.

5.5 Graph-Based Risk Propagation

Academic literature on financial fraud detection has explored graph-based approaches in which risk is propagated from known-bad nodes through transaction graphs to their neighbors. Applied to blockchain data, graph-based methods can identify wallets that are structurally connected to known exploit contracts even if they have not been directly implicated.

Graph propagation approaches represent a meaningful advance over flat blacklisting, but they remain reactive: they require at least one confirmed bad node as a seed, and their effectiveness is bounded by the density of the transaction graph. Entities that operate through intermediary contracts or that structure their attack staging to avoid direct graph adjacency to known-bad addresses are difficult to score accurately with graph methods alone.

5.6 Synthesis

The survey of prior approaches reveals a consistent pattern: each class of system optimizes for a specific signal type or threat model while remaining structurally blind to the dimensions it does not address. Blacklists fail against novel attacks. Simulation fails against temporal exploits. Rule-based systems fail against adversarial adaptation. Single-chain identity systems fail in multi-chain environments. Graph propagation fails against isolated or staged attacks.

The Cencera Trust Protocol is designed around the observation that no single signal domain is sufficient, and that trust must be treated as a longitudinal, multi-signal, probabilistic quantity rather than a binary attribute or a point-in-time assessment. The four signal domains described in Section 4 are chosen specifically to address the failure modes identified above: on-chain behavioral history addresses temporal exploitation, market signals address liquidity manipulation, off-chain signals address identity continuity, and AI-derived signals address adversarial adaptation through bytecode similarity analysis and behavioral baseline modeling.

6. BUSINESS MODEL & ECONOMICS

6.1 Pricing Structure

Call Type	Price Range	Gross Margin
Cached Entity Trust Query	$0.0003 – $0.0008	~96%
Transaction Risk Context	$0.001 – $0.003	~95%
Batch Screening (per entity)	$0.0002 – $0.0005	~96%
Historical Reputation Query	$0.01 – $0.05	~90%
Fresh Scoring Run	Blended	~75%
Enterprise Contract	$10K – $30K / month	73–87%

The primary variable cost is blockchain RPC data for fresh scoring runs (~$0.00015–$0.0005/call), not AI inference (~$0.000003/call). The engineering priority is maximizing cache coverage so that >90% of production calls are cache hits (serving cost <$0.00002/call), driving blended gross margins well above 90% at scale.

6.2 Go-To-Market Strategy

CENCERA's go-to-market strategy prioritizes enterprise contracts before public API volume. A single $15,000/month enterprise agreement covers months of infrastructure cost and provides anchor-customer validation for subsequent enterprise sales cycles. Three to five signed enterprise partners prior to public launch is the primary initial objective.

- Priority: DeFi protocol integrations via enterprise contracts ($10K–$30K/month).
- Secondary: Wallet provider integrations (Phantom, MetaMask) — high-volume recurring API revenue.
- Tertiary: Public API — developer adoption, community trust-building, long-tail volume.

6.3 Volume Milestones & Margin Profile

Monthly API Volume	Estimated Revenue	Net Operating Margin
< 50M calls	< $25K	Negative (burn phase)
50M – 200M calls	$25K – $100K	Near breakeven
200M – 500M calls	$100K – $250K	20 – 40%
500M – 1B calls	$250K – $500K	45 – 60%

Monthly API Volume	Estimated Revenue	Net Operating Margin
> 1B calls	> $500K / month	60 – 75%+

7. PLANNED EXTENSIONS & ROADMAP

Zero-Knowledge Trust Proofs

Cryptographic proofs that an entity's score meets a specified threshold, without revealing the underlying score or signals. Enables privacy-preserving trust gating for applications handling sensitive user data.

Community Signal Contributions

A governed mechanism through which community participants contribute verified behavioral signals, with staking-based incentives for accurate contributions and slashing penalties for false reports.

On-Chain Trust Attestations

Smart contract-readable attestations enabling composability without an API call at execution time — allowing DeFi protocol logic to reference trust scores natively in Solidity.

Institutional Reporting Module

Regulatory-grade, auditable trust reporting for institutional platforms operating under AML/KYC compliance frameworks.

CenceraAI Mainnet Deployment

Migration of the AI Agent infrastructure from BNB Testnet to BNB Mainnet following stability validation and comprehensive security audit.

Expanded Agent Network

CenceraAI integration with additional Agentverse agents to enable automated trust-scoring queries initiated directly by AI agents — positioning CENCERA as active infrastructure in the decentralized AI economy.

8. GLOSSARY

Term	Definition
CPID	Cencera Persistent Identifier — stable internal ID for each entity; persists across chain reorgs and contract upgrades.
Membase	Unibase's decentralized memory layer used by CenceraAI for persistent, cross-platform memory storage.
Innovation Score	On-chain metric that increments each time CenceraAI updates its memory on BNB Chain; a public, quantitative measure of the agent's cumulative growth.
Trust Manifold	The structured score object containing all metadata returned by the Trust Protocol API — composite score, trajectory, signals, and computation metadata.
Novel Behavior Flag	AI-generated signal when no known pattern matches a detected behavior; routes entity to human review with a provisional score.
Nascent Entity	An entity < 72 hrs old or < 50 interactions; receives a Limited Score with clearly labeled low confidence. Cold Start Pathway applies.
Score Trajectory	The five most recent score computation points for an entity — indicates whether the score is stable, improving, or deteriorating.
Cold Start Pathway	Dedicated scoring pipeline for Nascent entities using bytecode analysis, deployer trust inheritance, and initialization transaction analysis.
Retroactive Rescore	Upon confirmation of a new exploit, all similar entities are automatically rescored within 4 hours.
Tiered Explainability	Public API returns category-level triggers; enterprise partners receive signal-level detail under contract — transparency without enabling adversarial gaming.

9. LEGAL DISCLAIMER

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CENCERA's trust scores are probabilistic assessments, not guarantees. A high trust score does not guarantee the safety of a blockchain entity; a low score does not guarantee malicious intent. Consumers of Cencera Trust Protocol data are responsible for their own risk management policies and decisions.

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